

Input Protection Circuit

Step 2: eFuse Design			
Minimum recommended number of devices	TPS1685x	1	
Number of TPS1685x devices selected by the user, N		3	
I_{TRIP_TOTAL} (A) [Targeted]	Targeted circuit-breaker threshold during steady state [Typically 1.1 times I_{OUT_MAX} . Recommended to keep below 22A*N]	55	
V_{REF} (V) [Targeted]	Targeted threshold for overcurrent & short-circuit protection and active current sharing	1	
R_{REF} (k Ω) [Calculated]	Calculated value of resistance on the IREF pin	40	Need to connect a 1000pF capacitor at the IREF pin
R_{REF} (k Ω) [Actual]	Enter the value of resistance on the IREF pin	40.2	
V_{REF} (V) [Actual]	Actual threshold for overcurrent & short-circuit protection and active current sharing	1.005	
R_{MON} (k Ω) [Calculated]	Calculated value of resistance on the IMON pin	994.2	Need to connect a 22pF capacitor across R_{MON}
R_{MON} (k Ω) [Actual]	Enter the value of resistance on the IMON pin	1000.0	
I_{TRIP_TOTAL} (A) [Actual]	Actual circuit-breaker threshold during steady state	54.7	
R_{ILIM} (k Ω) [Calculated]	Calculated value of resistance on the ILM pin	1100.0	
R_{ILIM} (k Ω) [Actual]	Enter the value of resistance on the ILM pin	1100.0	
ILIM ACS (A) [Actual]	Actual active current sharing threshold. Recommended to keep below 20A	15.3	
T_{IMER} (ms) [Targeted]	Targeted transient overcurrent blanking interval	2	
C_{TIMER} (nF) [Calculated]	Calculated value of capacitance on the timer pin	2.6	
C_{TIMER} (nF) [Actual]	Enter the value of capacitance on the timer pin	3.3	
T_{IMER} (ms) [Actual]	Actual transient overcurrent blanking interval	3	
T_{SS} (ms) [Targeted]	Targeted startup time with nominal input voltage	60	
C_{OVRT} (nF) [Calculated]	Calculated value of the capacitor connected at the DVDT pin	60	
C_{OVRT} (nF) [Actual]	Enter the value of the capacitor connected at the DVDT pin	56	
T_{SS} (ms) [Actual]	Actual startup time with nominal input voltage	56	
I_{INRUSH} (A) [Calculated]	Total inrush current to charge the output capacitor. It is recommended to limit inrush current below 0.56A*N. N is total devices in parallel	1.7	
R_{UPPER} TO SET UVLO THRESHOLD (k Ω) [Actual]	Enter the value of upper resistor in a resistor divider network to set the under-voltage lockout threshold	1000	
R_{LOWER} TO SET UVLO THRESHOLD (k Ω) [Calculated]	Calculated value of lower resistor in a resistor divider network to set the under-voltage lockout threshold	30.9	
R_{LOWER} TO SET UVLO THRESHOLD (k Ω) [Actual]	Enter the value of lower resistor in a resistor divider network to set the under-voltage lockout threshold	31.0	
UVLO THRESHOLD (V) [Actual]	Actual under-voltage lockout threshold	39.9	
R_{UPPER} TO SET OVLO THRESHOLD (k Ω) [Actual]	Enter the value of upper resistor in a resistor divider network to set the over-voltage lockout threshold	1000	
R_{LOWER} TO SET OVLO THRESHOLD (k Ω) [Calculated]	Calculated value of lower resistor in a resistor divider network to set the over-voltage lockout threshold	19.8	
R_{LOWER} TO SET OVLO THRESHOLD (k Ω) [Actual]	Enter the value of the lower resistor in a resistor divider network to set the over-voltage lockout threshold	20.0	
OVLO THRESHOLD (V) [Actual]	Actual over-voltage lockout threshold	59.4	

8.4.2 Layout Example

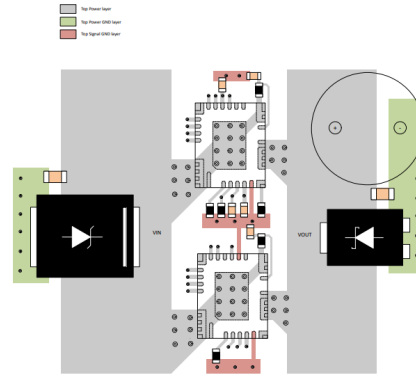
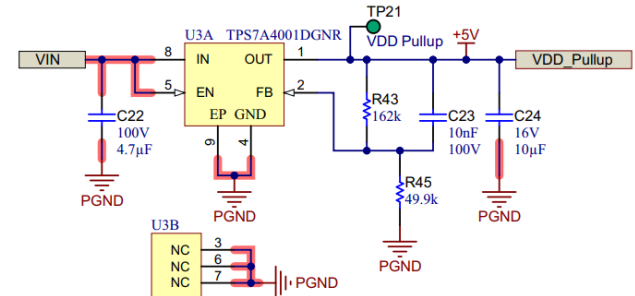
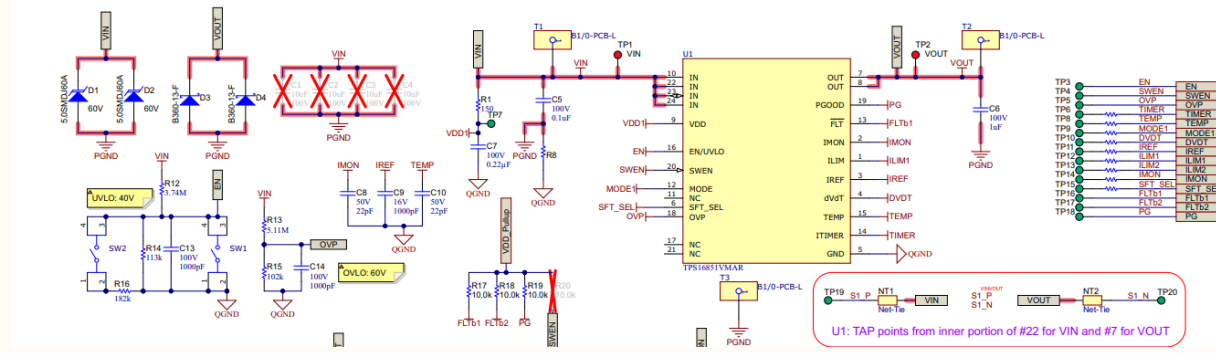
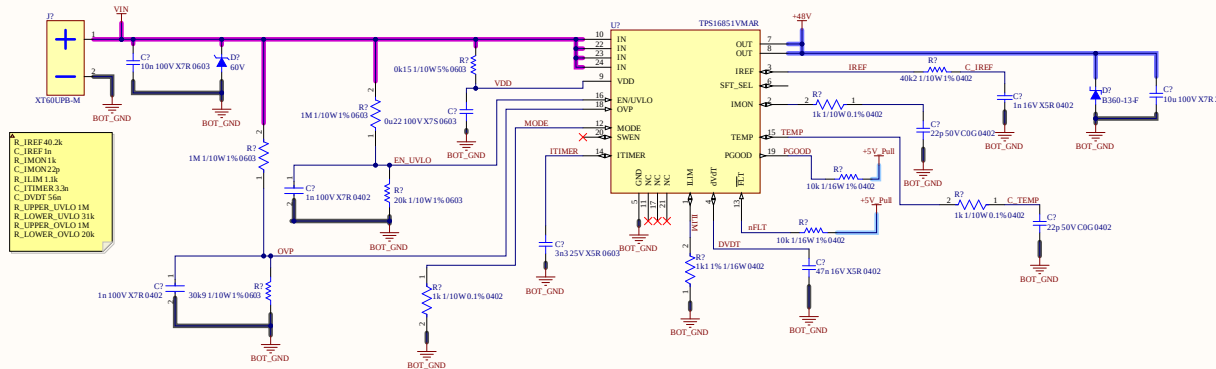


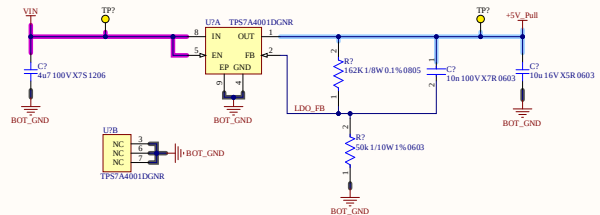
Figure 8-11. TPS1685x Two Parallel Devices Layout Example



eFuse 60V to 40V



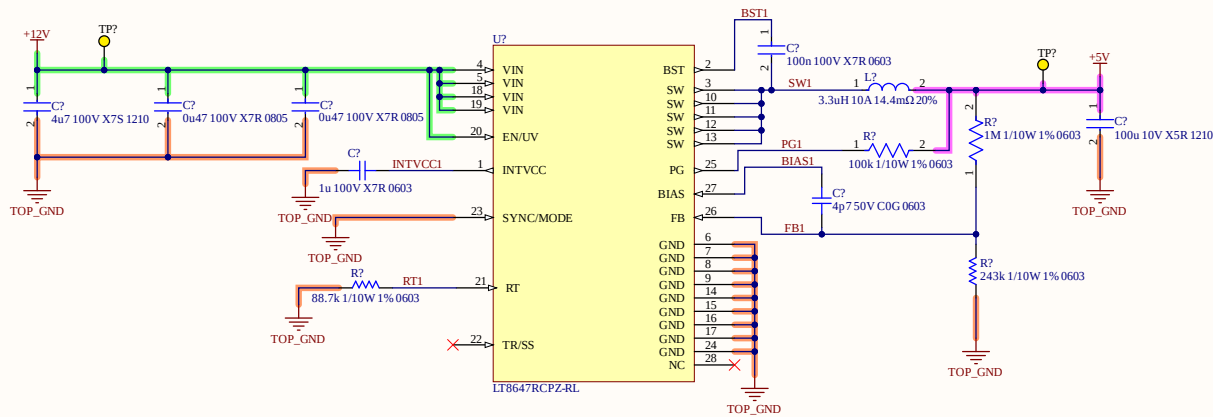
LDO 48V to 5V



Title: Protection Circuit		
Size: A2	Number:	Revision: 1.01
Date: 7/22/2008	Sheet: 1 of 1	Drawn By: Max Lan
File: C:\Users\j\Documents\Protection_Circuit_SchDoc	Sheet: 1 of 1	Drawn By: Max Lan

Cascaded Buck Power Stages

12V to 5V, 5A



48V to 12V, 13A

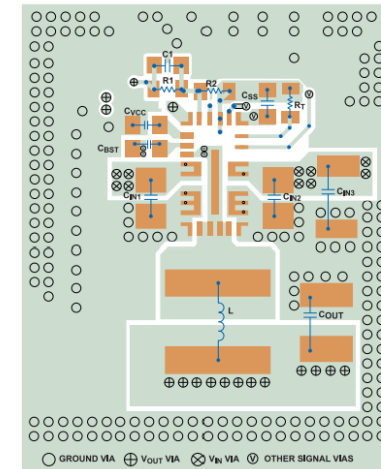
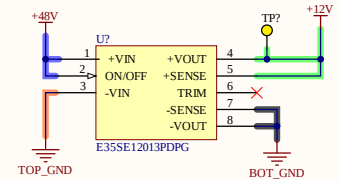


Figure 48. Recommended PCB Layouts for the LT8647

48V to 24V, 5A

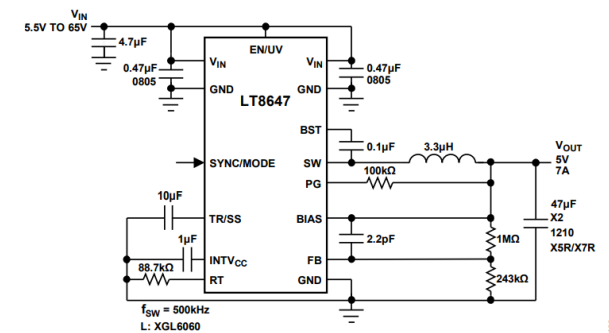
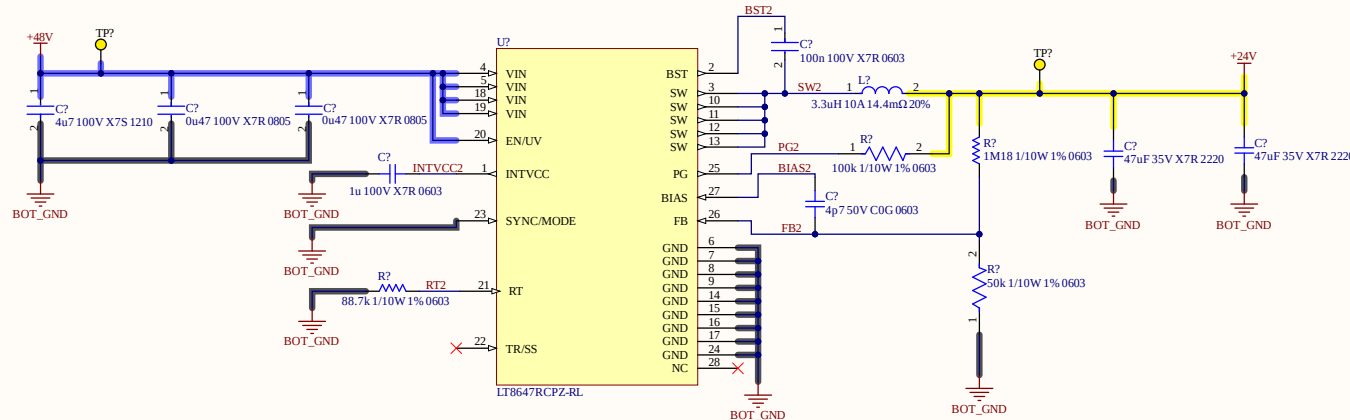
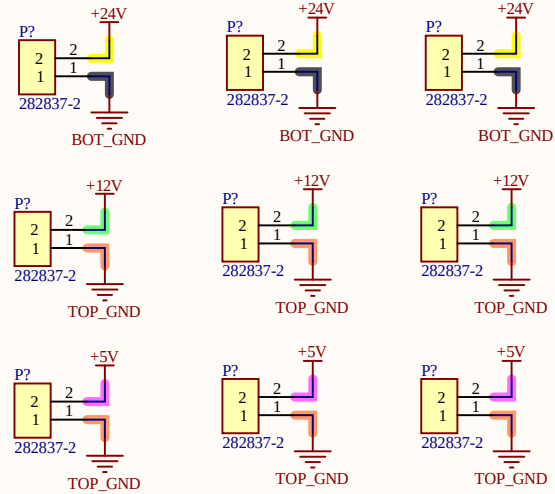
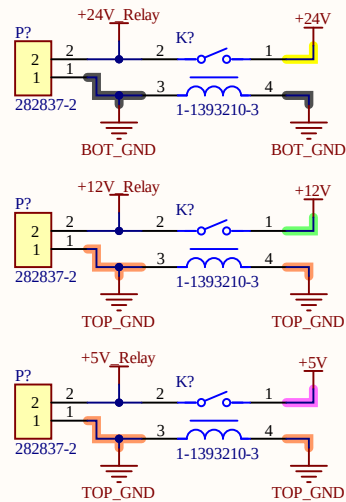


Figure 54. 5V, 7A Step-Down Converter with Soft-Start and Power Good

Title			
Power Stage Circuit			
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File:	C:\Users\...\Power_Stage.SchDoc	Drawn By: Max Lan	

Connectors

Use three relays for the connectors, one side with no relay, and another one with a relay that can sustain up to around 15-20A and a maximum of 30V



Title Connectors		
Size A4	Number	Revision 1.01
Date:	7/22/2026	Sheet of
File:	C:\Users\...\Connectors.SchDoc	Drawn By: Max Lan

[illegible]